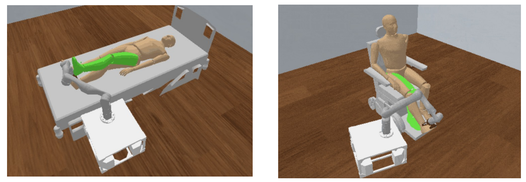
Learning to Accelerate Task and Motion Planning for Forceful Human Limb Manipulation

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logo

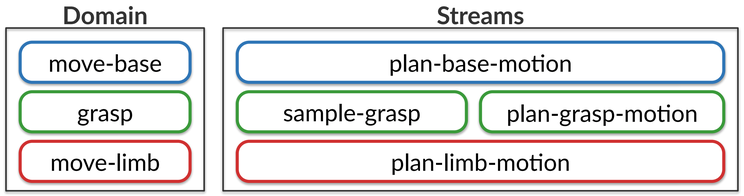
Motivation: Limb Manipulation

- Caregiving and rehabilitation using a robot require moving a person's limb by direct, forceful contact.
- Long-horizon: where to stand, where to grasp, how to move the limb ⇒ Task and Motion Planning (TAMP).
- The human's **range of motion (RoM) differs across people**



Problem Formulation

TAMP domain (PDDLStream): The plan-limb-motion stream is an MPC that searches over robot torques under the human and robot limits.



Task. $\tau = (q_h^0, q_r^0, q_h^{goal}) \sim P(\tau)$: move the limb to a goal configuration.

Objective. Reach the goal quickly without violating the human RoM.

Research Question

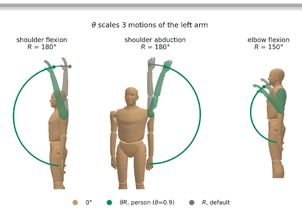
Learn the human's parameters θ
What is the human's range of motion?

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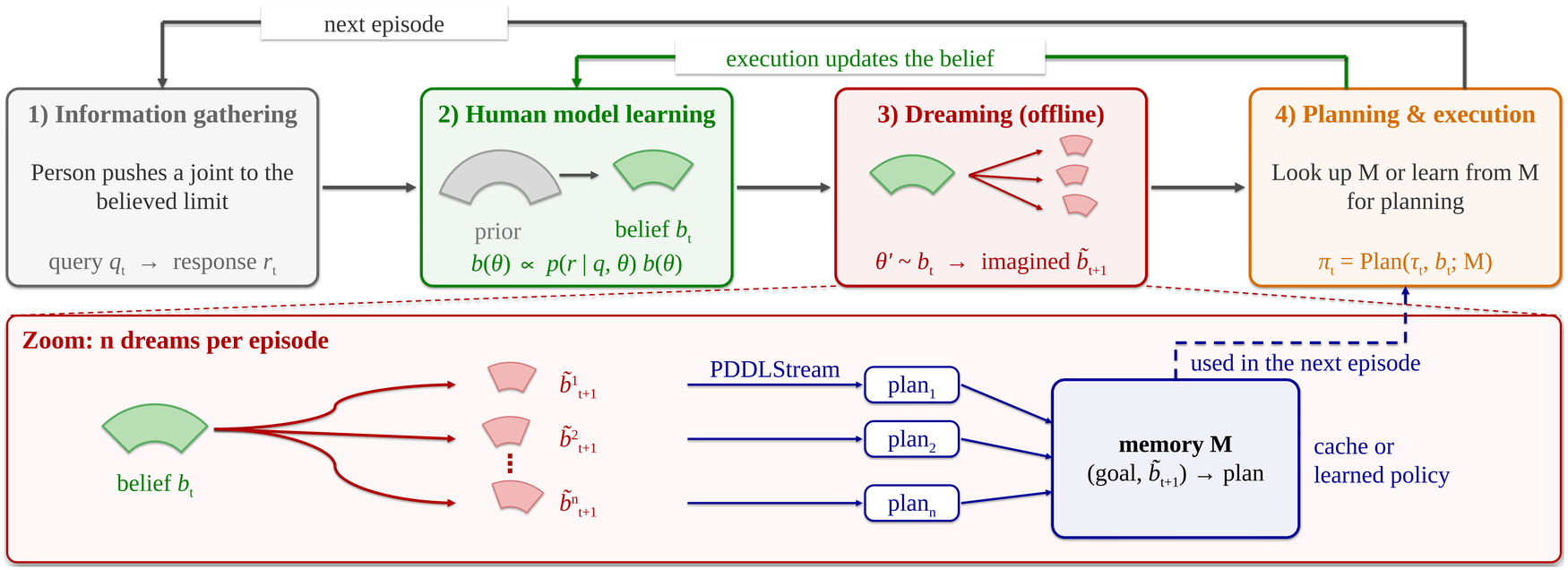
Learn the TAMP solver's parameters ψ
How can it plan fast for this person?

How can we learn to plan faster for the new user?

Human parameter θ: a person-specific range of motion



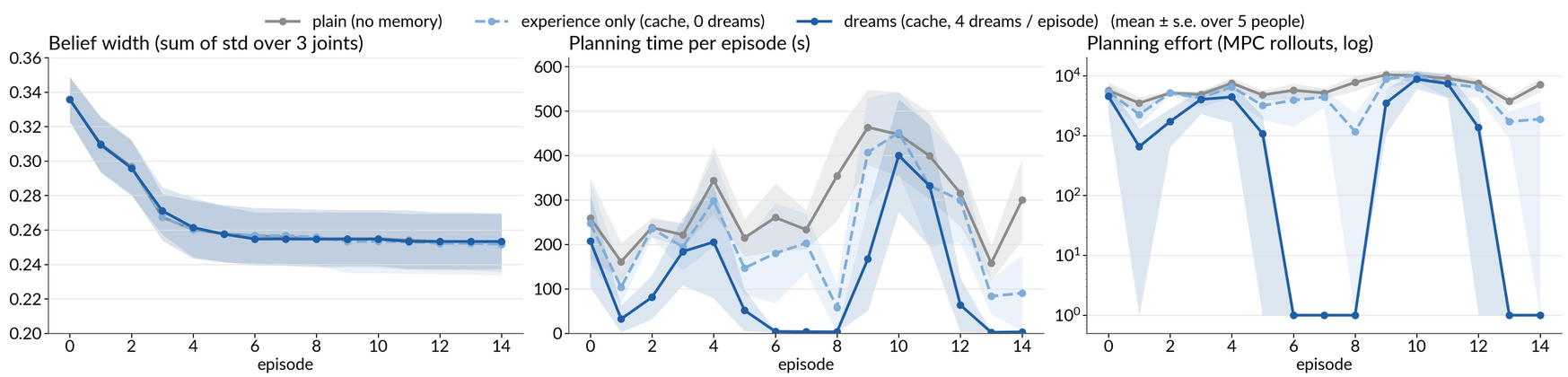
Approach



Experimental setup

- 5 simulated people × 15 episodes · left arm (3 joints) · 4 dreams / episode · 600 s per plan
- Tested on a left-arm manipulation environment in PyBullet sim and TidyBot
- **Evaluated: dreaming + planning & execution only** (no information gathering or human model learning steps)

Proof of concept



- **Dreams make planning a lookup:** median 4992 → **1** MPC rollouts, 223 s → **4 s** (plain → dreams).
- **Experience alone helps little:** median 196 s; 38 vs. **69** memory hits out of 75 episodes.
- **The belief narrows the same in all conditions** (0.34 → 0.25): memory speeds up planning, not learning.